

Name:

Surname:

Matricola:

Module 1:

Exercise 1) [10 points]

The temperature T of a membrane of area $S = 2$ is a function of the external temperature T_e and of the heating exchanged with a pump $n(t)$ as defined by the following.

$$\frac{dT(t)}{dt} = 0.4(T_e(t) - T(t)) + n(t)/S$$

the exchanged heating is imposed by a tension ruled by a selector percentage $v(t)$

$$n(t) = 200 v(t)$$

We are interested in the temperature $T(t)$.

1. Determine the dynamic equations that define the system.
2. Classify the system.
3. Find an equilibrium point such that the valve opening is 10 %.
4. If the equilibrium point exists, determine, if possible, its stability.

Exercise 2) [11 points]

Given the following LTI System

$$\begin{cases} \dot{x}_1(t) = -x_1(t) - 3x_2(t) + u(t) \\ \dot{x}_2(t) = x_1(t) - x_2(t) - u(t) \\ y(t) = x_1(t) + x_2(t) \end{cases}$$

1. Define the transfer function.
2. Is the transfer function a solid approximation of the system? Motivate the answer.

Exercise 3) [11 points]

Consider the transfer function:

$$G(s) = \frac{100}{s^2 + 10.1s + 1}$$

1. determine the following indexes for the response to a 3 unit step
 - a. Equilibrium value
 - b. Settling time at 5%
2. Is it possible to have overshoot?
3. Evaluate the response to the following signal: $\cos(100t) + \cos(t/100)$

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Module 2:

Exercise 1) [8 points]

Consider the transfer function:

$$G(s) = \frac{100(s^2 + 1)}{s^2 - 4}$$

1. Draw the root locus
2. Find the gain such that the closed loop system has a pole in the origin

Exercise 2) [11 points for all requests, 8 points for 2 requests]

Given the following system

$$G(s) = \frac{2s}{(s + 100)(s + 1)}$$

Define a controller in a nominal scenario such that the controlled output step response addresses at least 2 of the following requests:

- Show no overshoot
- Has an error lower than 5%
- Has a setting time of at most 10 seconds

Exercise 3) [13 points for all requests, 9 points for 3 requests]

Given the following system

$$G(s) = \frac{100}{s(10s + 1)(s + 100)}$$

Define a controller in a real scenario such that the controlled output step response addresses at least 3 of the following requests:

- Show an overshoot lower than 15%;
- Has an error lower than 5%;
- Has a setting time of at most 10 seconds;
- The measurement noise has an attenuation of at least 20 db.

Knowing that

- The measurement noise shows significant components only for frequency greater than 10KHz
- There is no disturbance

Exercise 1) [8 points]

Consider the transfer function:

$$G(s) = \frac{100(s^2 + 1)}{s^2 - 4}$$

1. Draw the root locus
2. Find the gain such that the closed loop system has a pole in the origin

$$G(s) = \frac{100(s^2 + 1)}{s^2 - 4}$$

• DRAW ROOT LOCUS

$$P = \{ -2; +2 \}$$

$$Z = \{ -j; +j \}$$

$$\sigma = \frac{P_1 + P_2}{2} = \frac{-2 + 2}{2} = 0 \rightarrow \text{NO ASYMPTOTE}$$

• FIND ORIGIN

$$\sigma_w = \frac{\sum P - \sum Z}{n} = \text{USELESS}$$

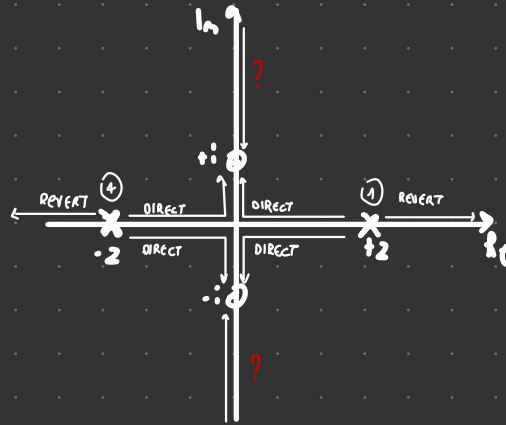
• FIND ANGLE

WITH $v=1$ IF WOULD BE $\left(\begin{array}{l} \text{TOO DIRECT} \\ \text{O. POWER} \end{array} \right)$

• GAIN TO HAVE POLE IN ORIGIN

DIRECT EDGE $\rightarrow$ POSITIVE GAIN

$$P = \frac{\prod m_i}{\prod \lambda_i} = \frac{2 \cdot 2}{1 \cdot 2} = 4$$



Exercise 2) [11 points for all requests, 8 points for 2 requests]

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$$G(s) = \frac{2s}{(s+100)(s+1)}$$

STABLE

• NO OVERSHOOT

• ERROR LOWER 5% $\longrightarrow e_{\infty} < 5\%$

• SETTING TIME AT MOST 10 SEC $\longrightarrow T_s \leq 10$

$$p = \{-100, -1\}$$

$$z = \{0\}$$

$$\bullet T_s \leq 10 \longrightarrow 2.2T \leq 10 \longrightarrow T \leq 4.54 \longrightarrow \text{CHOOSE } T=4$$

$$\bullet \text{ERROR } e_{\infty} < 5\% \longrightarrow |N-1| < 0.05 \longrightarrow -0.05 < N-1 < 0.05 \longrightarrow 0.95 < N < 1.05 \longrightarrow N=1 \text{ IS A GOOD CHOICE}$$

$$G_{OL}(s) = \frac{N}{Ts+1} = \frac{1}{4s+1}$$

IN THIS CASE IS $G_{OL}(s)$

$$C(s) = \frac{G_{OL}(s)}{G(s)} = \frac{1}{4s+1} \cdot \frac{(s+100)(s+1)}{2s}$$

NO OVERSHOOT SINCE $L = L \cdot G(s)$ IS FIRST ORDER
THERE IS NO OVERSHOOT

• $C(s)$ IS

• FEASIBLE $\text{DEG } N \leq \text{DEG } D$

• STABLE $\forall p \leq 0$

$$\longrightarrow \text{EX: } C(s) = \frac{1}{4s+1} \cdot \frac{(s+100)(s+1)}{1} \text{ NOT FEASIBLE } \because \text{DEG}(N) > \text{DEG}(D)$$

(ADD A NEW POLE ONE DECADE AFTER THE BIGGEST FREQUENCY)

$$C(s) = \frac{1}{4s+1} \cdot \frac{(s+100)(s+1)}{1} \cdot \frac{1}{s/100 + 1}$$

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Kowing that

- The measurement noise shows significant components only for frequency greater that 10Khz
- The is no disturbance

$$G(s) = \frac{100}{s(10s+1)(s+100)}$$

- OVERSHOOT LOWER THAN 15% $\rightarrow \xi < 15\%$
- ERROR LOWER 5% $\rightarrow e_{\infty} \leq 5\%$

• SETTING TIME AT MOST 10 SEC $\rightarrow T_s \leq 10 \text{ SEC}$

① TYPE

$\nu = 1 \rightarrow R_1 = N$

② OVERSHOOT CONSTRAINT

$\varphi_m > 5 \cdot 100 \rightarrow \varphi_m > 15^\circ$

③ ERROR CONSTRAINT $e_{\infty} \leq 5\%$

BUT SINCE TYPE 1 $\rightarrow S_0 \quad e_{\infty} = 0 \quad \ddot{u}$ THE END

④ FIND ω_c

• $2.2T_s \leq 10 \rightarrow T_s \leq 4.5 \rightarrow \frac{1}{\omega_c} \leq 4.5 \rightarrow \omega_c > \frac{1}{4.5} \text{ Hz}$

• $\omega_c \leq 10^4 \text{ Hz}$

SO $\rightarrow \frac{1}{4.5} \text{ Hz} < \omega_c < 10^4 \text{ Hz} \rightarrow \text{CHOOSE } \omega_c = 10^1 \text{ Hz}$

$\frac{20}{6} < 20$

⑤ CHECK IF ω_c CHOOSSED SATISFY CONSTRAINT

• $L(s) = R_1 G(s) = \frac{100N}{s(10s+1)(s+100)}$

• $\varphi_m > 15^\circ$

• $\varphi_m = 180 - |\Delta L(\omega_c)| \quad \omega_c = 10^1$

$\uparrow$

$\tan^{-1}(0/100) - \tan^{-1}(10/1) - \tan^{-1}(10 \cdot 10/1) - \tan^{-1}(10/100)$

$= 0 - 90 - 90 - 0 = -180$

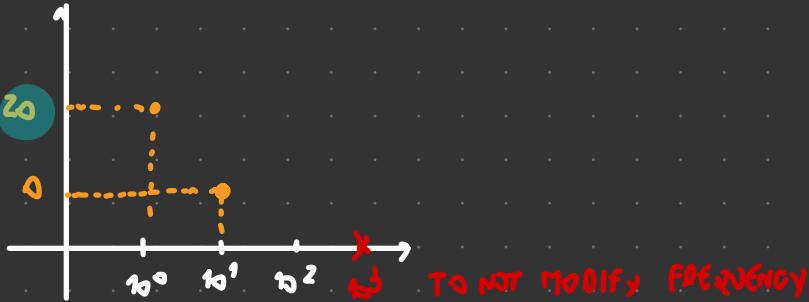
$\varphi_m = 180 - |-180| = 0$

NOT SATISFIED $\ddot{u}$

⑥ CHOOSE R_2 AND FIND L

• $R_2 = \frac{(10s+1)(s+100)}{\pi(1+T_s)}$

$L = R_1 R_2 G(s) = \frac{100N}{s(10s+1)(s+100)} \frac{(10s+1)(s+100)}{\pi(1+T_s)^2} = \frac{100N}{s(1+s/10^3)^2}$



⑦ FIND N

N SUCH THAT

$[10^0; 20 \log_{10}(N)]$

$20 \log_{10}(N) = 20$

$\downarrow$

$N = 10$

$\rightarrow 10 = 100N \rightarrow N = 1/10$

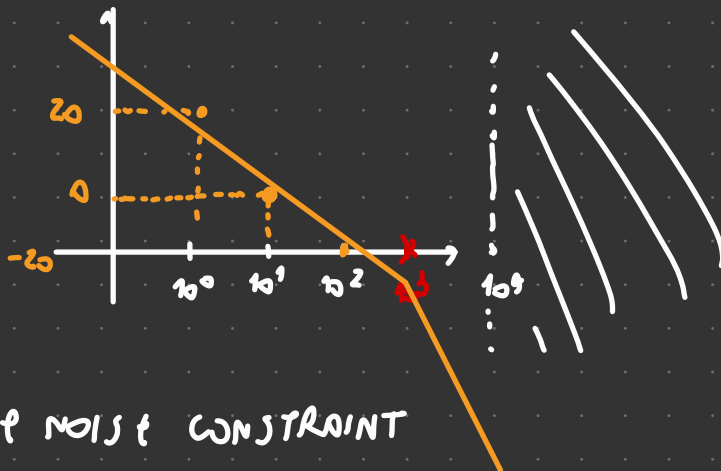
⑧ COMPUTE L

$$L = R_1 R_2 G(s) = \frac{100\mu}{s(11.3/\mu s)^2}$$

- FEASIBLE $\checkmark$
- STABLE $\checkmark$
- $\gamma_m > 15$? COMPUTE IT

NOISE CONSTRAINT

- $\omega_N = 20 \text{ kHz} = 10^4 \text{ Hz}$
- NOISE = 20 dB



SINCE I REACH 10^4 WITH
LESS THAN -20 dB I SATISFY THE NOISE CONSTRAINT

$$S(\omega) \approx L(\omega) \rightarrow < -20 \text{ dB}$$